Circuit Connections

Here are the precise connections based on your code.

1. Receiver (The Car) - Arduino Uno + L298N + HC-05

L298N Pin	Connected To	Description
ENA / ENB	**Jumper Caps**	Keep black caps ON (Enables full speed).
IN1	**Arduino Pin 8**	Left Motor Logic 1
IN2	**Arduino Pin 9**	Left Motor Logic 2
IN3	**Arduino Pin 10**	Right Motor Logic 1
IN4	**Arduino Pin 11**	Right Motor Logic 2
12V Input	**Battery (+)**	Red wire from 7.4V (Li-ion) battery pack.
GND	**Battery (-)**	**CRITICAL:** Connect Battery Black wire AND Arduino GND here.
5V Output	**Arduino Vin**	Powers the Arduino (No USB needed).
OUT 1 & 2	**Left Motors**	Connect to left side motor wires.
OUT 3 & 4	**Right Motors**	Connect to right side motor wires.

HC-05 Wiring Table

HC-05 Pin	Connected To	Description
VCC	**Arduino 5V**	Power (Module has internal regulator).
GND	**Arduino GND**	Common Ground.
TX	**Arduino Pin 0 (RX)**	Sends data *from* Bluetooth *to* Arduino.
RX	**Arduino Pin 1 (TX)**	Receives data *from* Arduino *to* Bluetooth.
State	**Not Connected**	Leave empty.
EN / Key	**Not Connected**	Leave empty (Used only for programming mode).

> ⚠️ **Important:** You must **unplug the HC-05 RX/TX wires** (Pin 0 and 1) while uploading the code to the Arduino. Reconnect them after the upload is done.

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2. Transmitter (The Glove) - ESP32 + MPU6050

MPU6050 Pin	ESP32 Pin	Description
VCC	**3.3V**	ESP32 usually outputs 3.3V logic.
GND	**GND**	Common ground.
SCL	**GPIO 22**	I2C Clock Line.
SDA	**GPIO 21**	I2C Data Line.
